THE LANCET
Digital Health

Supplementary appendix 1

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

Supplement to: Vaidya P, Bera K, Gupta A, et al. CT derived radiomic score for predicting the added benefit of adjuvant chemotherapy following surgery in stage I, II resectable non-small cell lung cancer: a retrospective multicohort study for outcome prediction. *Lancet Digital Health* 2020; published online Feb 13. https://doi.org/10.1016/S2589-7500(20)30002-9.

Appendix-1

Section 1: Radiomic Feature Extraction.

Lung nodules were annotated by an expert radiologist using 3D-slicer (3D Slicer, version 4.6: NIH-funded; https://www.slicer.org) and a freehand tool. Intratumoral and peritumoral area was selected using these annotations. For intratumoral region, the annotated tumor is used as the area of interest. Following annotations, the peritumoral region corresponding to the lesion was defined using the following steps. Firstly, a morphological operation of dilation was performed to capture the area outside the nodule, up to a radial distance of 15mm from the lesion boundary. The choice of peri-nodular size was based on previous findings, where a resection margin >15mm did not have a prognostic effect in the context of disease recurrence. The intranodular mask was then subtracted from this dilated mask to obtain a ring of lung parenchyma immediately around the nodule. The perinodular region was subsequently divided into five equally spaced 3-mm rings.

Radiomic features were extracted from the three slices having the maximum area of the tumor using software developed in the Center of Computational Imaging and Personalized Diagnostics, Case Western Reserve University, implemented on a MATLAB release 2016a platform (MathWorks, Natick, MA, USA).

A total of 124 textural descriptors were commutated voxel-wise across the three slices from each of the annotated region. The features included 48 from Gabor feature family, 13 from Haralick, 13 from Collage, 25 from Laws, and 25 from Laplace feature families. The First-order statistics (mean, median, SD, skewness, and kurtosis) were calculated from each feature vector. Table-S1 explains all the extracted features.

Total Extracted Radiomic Features.

Table S.1 – Total Radiomic Features extracted from intratumoral and annular ring-shaped peritumoral regions. A total of 6 statistics- (mean, median, kurtosis, skewness, standard deviation, and range) were calculated for each feature.

Radiomic Feature Family	Radiomic Feature Parameters	Radiomic Feature Family	Radiomic Feature Parameters
Haralick Feature Family	Entropy, Window Size = 5	**Laws Energy Feature Family**	Level, Level
	Energy, Window Size = 5		Level, Edge
	Inertia, Window Size = 5		Level, Spot
	Inverse Difference Moment , Window Size = 5		Level, Wave
	Correlation, Window Size = 5		Level, Ripple
	Information Measure of Correlation-1, Window Size = 5		Edge, Level
	Information Measure of Correlation-2, Window Size = 5		Edge, Edge
	Sum Average, Window Size = 5		Edge, Spot
	Sum Variance, Window Size = 5		Edge, Wave
	Sum Entropy, Window Size = 5		Edge, Ripple
	Difference Average, Window Size = 5		Spot, Level
	Difference Variance, Window Size = 5		Spot, Edge
	Difference Entropy, Window Size = 5		Spot, Spot
Gabor Wavelet Feature Family	Orientation=0, Bandwidth=1, Frequency=0		Spot, Wave
	Orientation=0, Bandwidth=1, Frequency=2		Spot, Ripple
	Orientation=0, Bandwidth=1, Frequency=4		Wave, Level
	Orientation=0, Bandwidth=1, Frequency=6		Wave, Edge
	Orientation=0, Bandwidth=1, Frequency=8		Wave, Spot
	Orientation=0, Bandwidth=1, Frequency=10		Wave, Wave
	Orientation=π/8, Bandwidth=1, Frequency=0		Wave, Ripple
	Orientation=π/8, Bandwidth=1, Frequency=2		Ripple, Level
	Orientation=π/8, Bandwidth=1, Frequency=4		Ripple, Edge
	Orientation=π/8, Bandwidth=1, Frequency=6		Ripple, Spot
	Orientation=π/8, Bandwidth=1, Frequency=8		Ripple, Wave
	Orientation=π/8, Bandwidth=1, Frequency=10		Ripple, Ripple
	Orientation=π/4, Bandwidth=1, Frequency=0	**Laplace Feature Family**	Level, Level
	Orientation=π/4, Bandwidth=1, Frequency=2		Level, Edge
	Orientation=π/4, Bandwidth=1, Frequency=4		Level, Spot
	Orientation=π/4, Bandwidth=1, Frequency=6		Level, Wave
	Orientation=π/4, Bandwidth=1, Frequency=8		Level, Ripple
	Orientation=π/4, Bandwidth=1, Frequency=10		Edge, Level

	Orientation=3π/8, Bandwidth=1, Frequency=0		Edge, Edge
	Orientation=3π/8, Bandwidth=1, Frequency=2		Edge, Spot
	Orientation=3π/8, Bandwidth=1, Frequency=4		Edge, Wave
	Orientation=3π/8, Bandwidth=1, Frequency=6		Edge, Ripple
	Orientation=3π/8, Bandwidth=1, Frequency=8		Spot, Level
	Orientation=3π/8, Bandwidth=1, Frequency=10		Spot, Edge
	Orientation=π/2, Bandwidth=1, Frequency=0		Spot, Spot
	Orientation=π/2, Bandwidth=1, Frequency=2		Spot, Wave
	Orientation=π/2, Bandwidth=1, Frequency=4		Spot, Ripple
	Orientation=π/2, Bandwidth=1, Frequency=6		Wave, Level
	Orientation=π/2, Bandwidth=1, Frequency=8		Wave, Edge
	Orientation=π/2, Bandwidth=1, Frequency=10		Wave, Spot
	Orientation=5π/8, Bandwidth=1, Frequency=0		Wave, Wave
	Orientation=5π/8, Bandwidth=1, Frequency=2		Wave, Ripple
	Orientation=5π/8, Bandwidth=1, Frequency=4		Ripple, Level
	Orientation=5π/8, Bandwidth=1, Frequency=6		Ripple, Edge
	Orientation=5π/8, Bandwidth=1, Frequency=8		Ripple, Spot
	Orientation=5π/8, Bandwidth=1, Frequency=10		Ripple, Wave
	Orientation=3π/4, Bandwidth=1, Frequency=0		Ripple, Ripple
	Orientation=3π/4, Bandwidth=1, Frequency=2		Entropy, Window Size = 5
	Orientation=3π/4, Bandwidth=1, Frequency=4		Energy, Window Size = 5
	Orientation=3π/4, Bandwidth=1, Frequency=6		Inertia, Window Size = 5
	Orientation=3π/4, Bandwidth=1, Frequency=8		Inverse Difference Moment , Window Size = 5
	Orientation=3π/4, Bandwidth=1, Frequency=10		Correlation, Window Size = 5
	Orientation=7π/8, Bandwidth=1, Frequency=0	**CoLIAGe Feature Family**	Information Measure of Correlation-1, Window Size = 5
	Orientation=7π/8, Bandwidth=1, Frequency=2		Information Measure of Correlation-2, Window Size = 5
	Orientation=7π/8, Bandwidth=1, Frequency=4		Sum Average, Window Size = 5
	Orientation=7π/8, Bandwidth=1, Frequency=6		Sum Variance, Window Size = 5
	Orientation=7π/8, Bandwidth=1, Frequency=8		Sum Entropy, Window Size = 5
	Orientation=7π/8, Bandwidth=1, Frequency=10		Difference Average, Window Size = 5
			Difference Variance, Window Size = 5
			Difference Entropy, Window Size = 5

Section 2: Radiomic Feature Stability.

Feature stability and reproducibility were evaluated using the RIDER test-retest dataset. This dataset had 31 lung cancer patients, scanned two times with 15 mins' difference apart. Two scans of every patient were used for calculating the intraclass correlation coefficient (ICC) for each feature vector. Considering the threshold of 0.85, the analysis was performed using all feature vectors having ICC values higher than the threshold limit. After completing the above experiment, a total of 757 features were retained from the intratumoral and peritumoral feature pool of 4464. These stable features were used for further analysis for feature selection and model building.

Section 3: CT Scan parameters.

Table S.2 – CT scan parameters as per the datasets

CCF			UPENN			TCIA		
	No. of studies	**Kernal**		**No. of studies**	**Kernal**		**No. of studies**	**Kernal**
Siemens	250	B30f, B30s, B31f, B31s, B35f, B35s, B40sf, B40s, B41f, B60f, B60s, B70f, B70s	**Siemens**	107	B30f, B31f, B31s, B40f, B41f, B50f, B60f, B70f, Br40d\3, Br59d\3, I30f\3, I31f\3, I40f\3, I40s, I50f\3, I70f\2, I80s\2, T20s	**Siemens**	13	B31f, B40f, B45f, B50f, B70f, I70f\2
Philips	38	B,C,D,L,YA,YB	**Phillips**	11	B, YB, E,	**Phillips**	2	YB, C
Toshiba	14	FC03, FC12, FC13, FC14, FC52, FC86	**Toshiba**	2	FC56, FC18	**Toshiba**	1	FC52
GE	22	Detail, Lung, SOFT, Standard	**GE**	44	Standard, Lung, CHST	**GE**	66	Standard, Lung, BONE, BONEPLUS, SOFT
Hitachi	4	04,09						
<=1	134		**<=1**	10		**<=1**	16	
<1<=3	205		**<1<=3**	33		**<1<=3**	59	
3< <=5	61		**3< <=5**	70		**3< <=5**	7	
With Contrast	65		**With contrast**	32		**With contrast**	22	
W/o Contrast	223		**W/o Contrast**	82		**W/o Contrast**	60	

Section 4: Radiomic Top Features.

Top features selected using LASSO Cox regression model. Considering alpha=0.1, top 13 features were selected, which gave the optimum C-index on training set (D_{1s}, N=256).

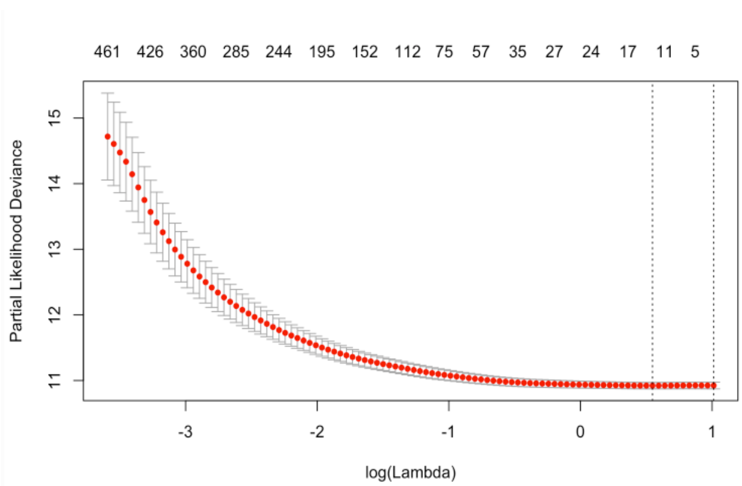


Figure S.1- Feature selection using the least absolute shrinkage and selection operator (LASSO) Cox model. Tuning parameter (λ) selection in the LASSO model used 10-fold cross-validation via minimum criteria. The area under the receiver operating characteristic (AUC) curve was plotted versus log(λ). Dotted vertical lines were drawn at the optimal values by using the minimum criteria and the 1 standard error of the minimum criteria. Optimal λ was 2.7528 which resulted in 13 nonzero coefficients.

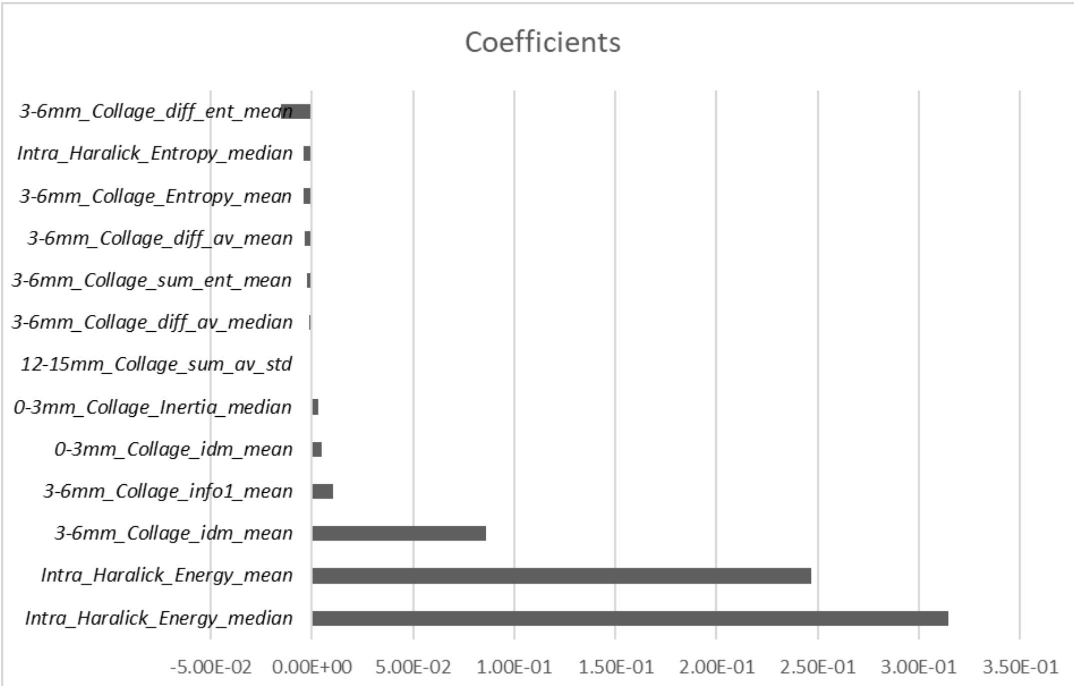


Figure S.2- LASSO Coefficients for top 13 selected Radiomic textural features.

Table S.3 - Top selected features from LASSO model along with its LASSO coefficient

	Feature Family	Descriptor	Location	Statistics	LASSO coefficient
1	Haralick	Energy	intratumoral	Median	3·14E-01
2	Haralick	Energy	Intratumoral	Mean	2·47E-01
3	CoLlAGe	idm	3-6mm	Mean	8·59E-02
4	CoLlAGe	Info1	3-6 mm	Mean	1.02E-02
5	CoLlAGe	idm	0-3 mm	Mean	4.80E-03
6	CoLlAGe	Inertia	0-3 mm	Median	2.95E-03
7	CoLlAGe	Sum_av	12-15 mm	Std	-4.15E-05
8	CoLlAGe	Diff_av	3-6 mm	Median	-1.62E-03
9	CoLlAGe	Sum_ent	3-6 mm	Mean	-2.35E-03
10	CoLlAGe	Diff_av	3-6 mm	Mean	-3.49E-03
11	CoLlAGe	Entropy	3-6 mm	Mean	-4.02E-03
12	Haralick	Entropy	Intratumoral	Median	-4.41E-03
13	CoLlAGe	Diff_ent	3-6 mm	Mean	-1.55E-02

Section 5: Development of QuRiS.

The QuRiS was developed using the formula below.

$$\textbf{QuRiS}=3.14\text{E-01}\times(\text{Intra\_Haralick\_Energy\_median}) + 2.47\text{E-01}\times(\text{Intra\_Haralick\_Energy\_mean}) + 8.59\text{E-02}\times(\text{3-6mm\_Collage\_idm\_mean}) + 1.02\text{E-02}\times(\text{3-6mm\_Collage\_info1\_mean}) + 4.80\text{E-03}\times(\text{0-3mm\_Collage\_idm\_mean}) + 2.95\text{E-03}\times(\text{0-3mm\_Collage\_Inertia\_median}) - 4.15\text{E-05}\times(\text{12-15mm\_Collage\_sum\_av\_std}) - 1.62\text{E-03}\times(\text{3-6mm\_Collage\_diff\_av\_median}) - 2.35\text{E-03}\times(\text{3-6mm\_Collage\_sum\_ent\_mean}) - 3.49\text{E-03}\times(\text{3-6mm\_Collage\_diff\_av\_mean}) - 4.02\text{E-03}\times(\text{3-6mm\_Collage\_Entropy\_mean}) - 4.41\text{E-03}\times(\text{Intra\_Haralick\_Entropy\_median}) - 1.55\text{E-02}\times(\text{3-6mm\_Collage\_diff\_ent\_mean})$$

Distribution of QuRiS across three datasets

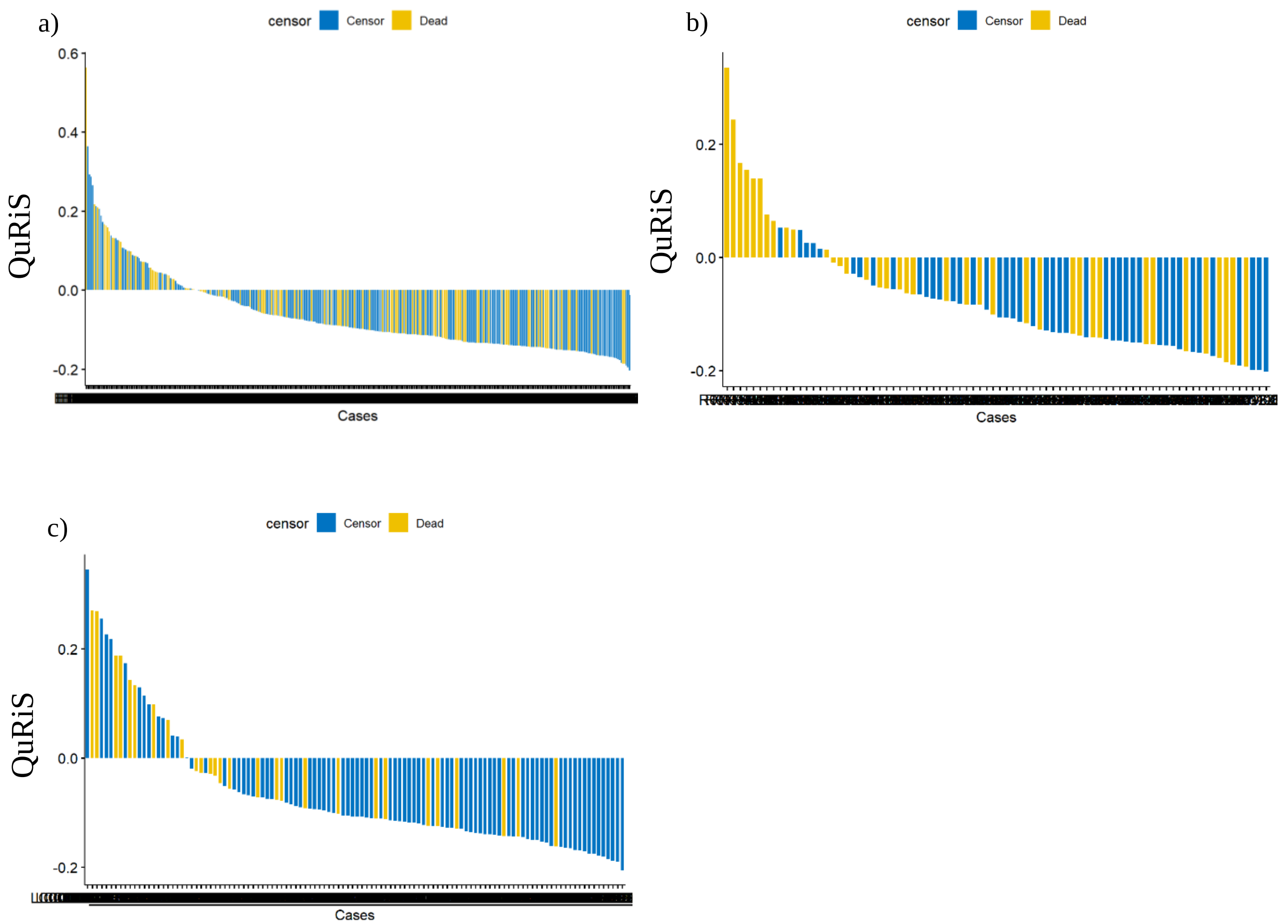


Figure S3- Distribution of QuRiS across three independent datasets a) D_1, b) D_2, and c) D_3.

Section 6: Selecting a threshold from QuRiS.

The best threshold for prognostic and predictive validation was selected using the training dataset D_1. Every point on the QuRiS was considered as a potential candidate for threshold. We excluded the extreme QuRiS values as possible thresholds that would create an extremely skewed group distribution.

For the purpose of prognosis prediction, we divided the patients into two potential groups based on each threshold value. The best threshold was selected having a maximum hazard ratio and consecutively, minimum p-value, and the maximum difference between the median survival between two predicted groups.
For the purposes of predicting the added benefit of adjuvant chemotherapy, a similar procedure was performed, but instead of two groups, the dataset was divided into three groups (Qh, Qi, QL). Within these groups, we compared patients who received ACT and patients who underwent surgery alone.

Section 7: Univariate Analysis.

The most prognostic clinical variables were selected using univariate analysis and hazard ratios using training cohort D_1. In the subset of patients who received surgery alone without any additional treatment, the following clinical factors were independently prognostic.

Patients who received surgery alone

Table S.4 - Univariate Analysis of clinical variables. Stage, T-, N- descriptor, LVI status were independently prognostic of DFS in the D_{1s} cohort where patients received surgery alone

		C-Index	HR	P-Value
1	AJCC Stage	0·56	2.59 [1.48-4.53]	<0.001
2	**T-Descriptor**	**0·58**	**2.77 [1.10-7.01]**	**0.02**
3	**N-Stage**	**0·53**	**4·3 [1.88-10.17]**	**<0.001**
4	**LVI**	**0·64**	**2·42 [1.53-3.82]**	**<0.001**
5	Margin Status	0.52	3.27 [1.19-8.97]	0.01
6	Tumor Size	0.58	1.00 [1.00-1.01]	0.01
7	Type of surgery	0.52	0.81 [0.59-1.13]	0.2
8	Histology	0·52	1·5 [0.63-3.89]	0.6
9	EGFR Mutation	0.52	0.81[0.34-1.89]	0.6

Patients who received surgery + ACT

The same prognostic factors were evaluated within the patient cohort who received chemotherapy following surgery. Developed QuRiS had a similar performance to that of these previously defined clinical biomarkers.

Table S.5 - Univariate Analysis of clinical variables in D_{1a} cohort where patients received ACT following surgery.

		C-Index	HR	P-Value
1	AJCC Stage	0·52	0.61 [0.25-1.47]	0.3
2	T-Stage	0·54	1·2 [0.34-4.86]	0.3
3	N-Stage	0·56	1.76 [0.86-3.59]	0.1
4	LVI	0·52	1.12 [0.56-2.26]	0.7
5	Margin Status	NA	NA	NA
6	Tumor Size	0.52	0.99 [0.97-1.01]	0.6
7	Type of surgery	0.53	1.36 [0.76-2.44]	0.3
8	Histology	0·53	0.68 [0.43-1.58]	0.3
9	EGFR Mutation	0.49	0.86 [0.31-2.38]	0.8

Section 8: Comparing QuRiS against clinical parameters.

The maximum correlation of QuRiS was observed with tumor size compared to all other clinical factors. Although the correlation was statistically significant, tumor size had hazard ratio 1 [1.00-1.01] for predicting DFS, whereas QuRiS had hazard ratio 1.5 [1.08-2.00].

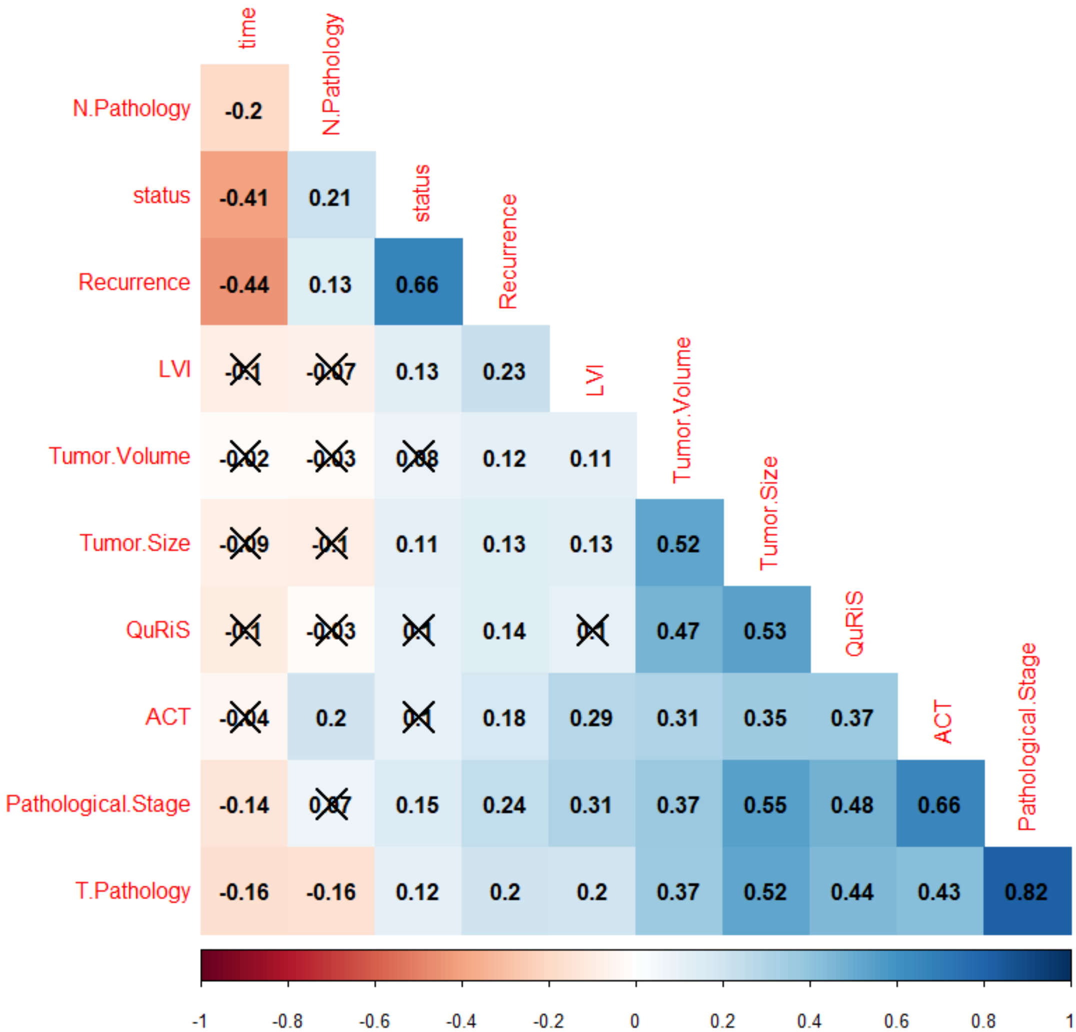


Figure S.4: Pearson correlation coefficient between constructed QuRiS and clinical factors.

Section 9: KM plots for stage I patients.

The patients from D_1, D_2, and D_3 were analyzed for QuRiS within stage I patients (IA, IB). In this clinically defined low-risk cohort, we could observe a clear difference between high risk and low-risk tumors using QuRiS.

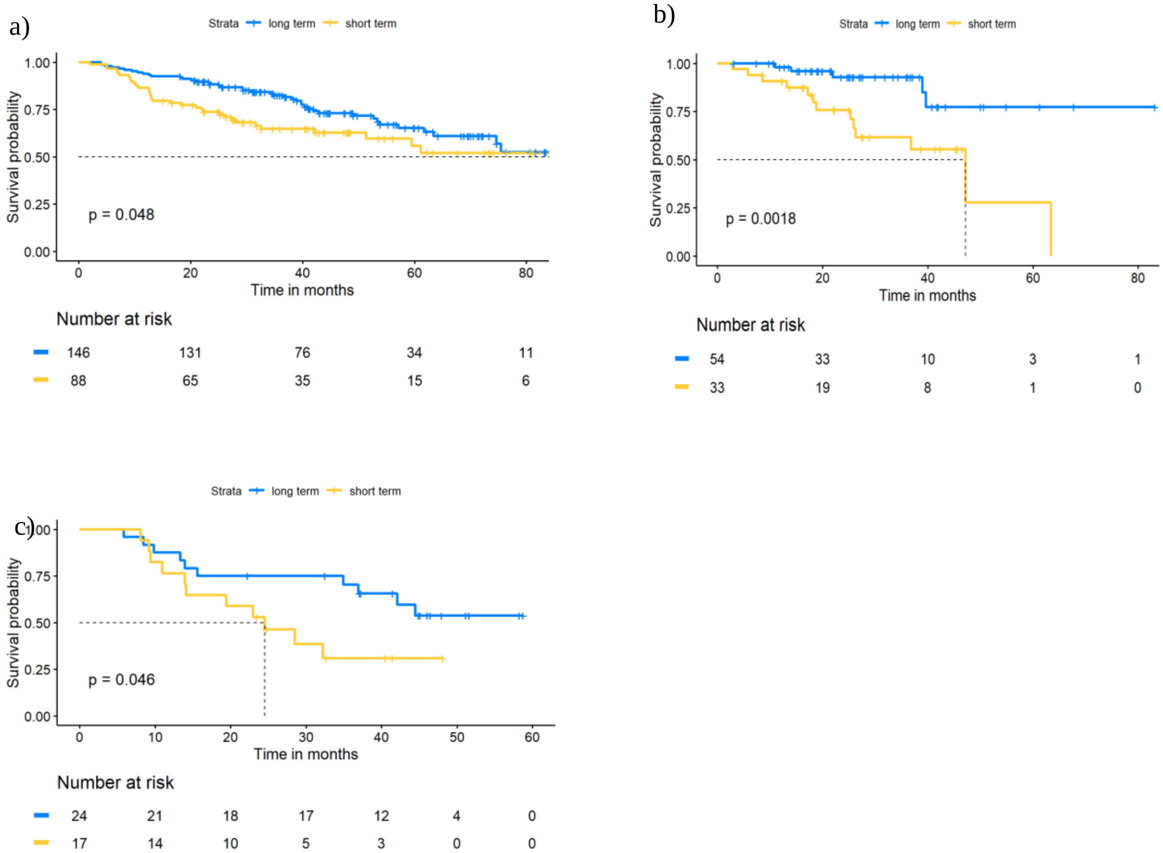


Figure S.5: Kaplan-Meier Survival Curves for a) Dataset D_1, b) Dataset D_2 and c) Dataset D_3 for Stage-I patients. The patients with high QuRiS had significantly worse survival compared to low QuRiS patients. In this clinically defined low-risk group, we could observe two clear groups for high-risk and low-risk groups.

Section 10: Prognostic and Predictive Performance of stage IB patients.

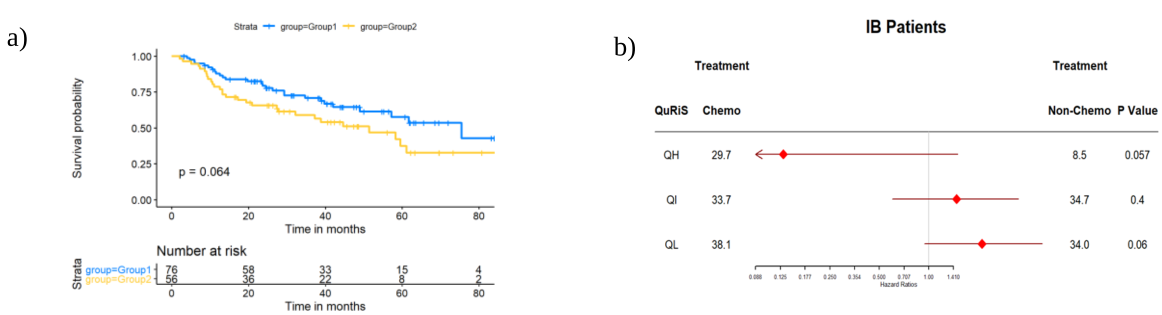


Figure S.5: a) Kaplan-Meier Survival Curves for IB patients. b) Predictive added benefit of ACT using developed QuRiS and IB patient cohort.

Section 11: Predictive performance of QuRiS.

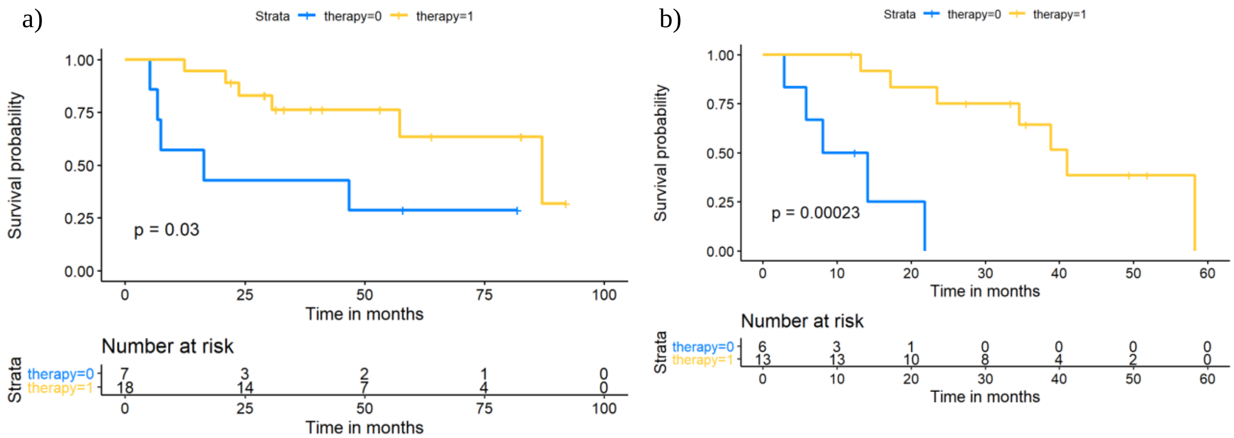
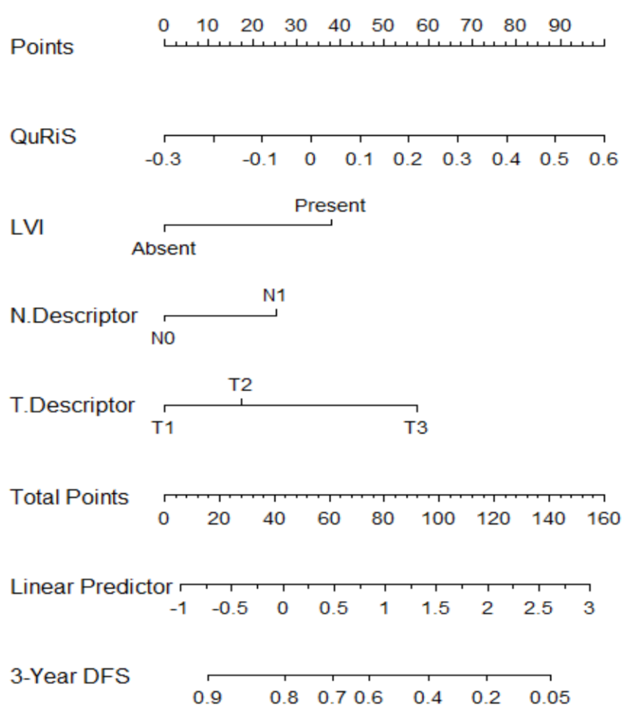


Figure S.6- Subset of patients from a) D_1 b) $D_2 + D_3$ with QuRiS greater than 0·098. The patients were analyzed based on the treatment received. In this cohort, patients who received ACT had significantly better survival than the patients who received surgery alone.

Section 12: Construction of QuRNom.



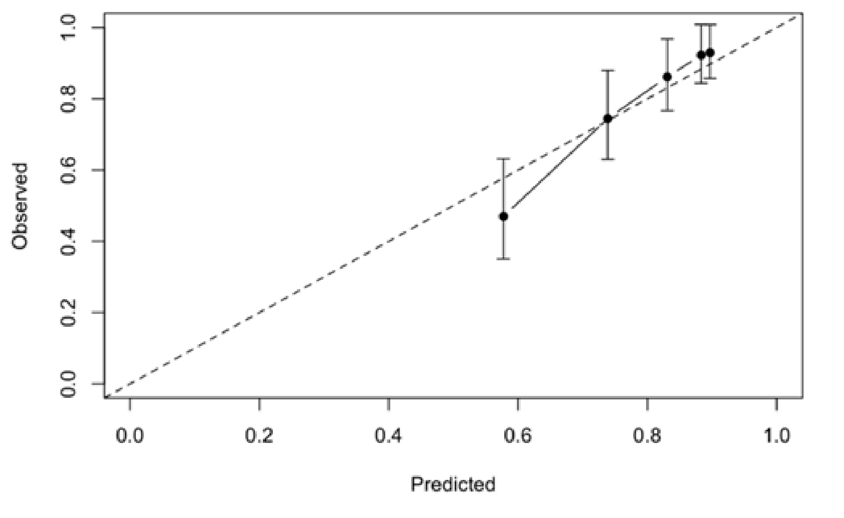


Figure S.7- Constructed QuRNom for stage I, II lung cancer along with the assessment of the model calibration. a) The nomogram was developed using D1s (N=257) using QuRiS, lymphovascular invasion, N-descriptor and T-descriptor. b) Calibration curve for QuRNom which depicts the calibration of the model in terms of the agreement between the predicted DFS and observed DFS. The y-axis represents the actual DFS and x-axis represents observed DFS. The black solid line represents the performance of the nomogram, of which a closer fit to the diagonal dotted line represents a better prediction.

The following parameters were involved for constructing QuRNom and had significant contribution for constructing QuRNom.

Table S.6 –Detailed scores of prognostic factors in the nomogram

Characteristics		**Nomogram**
LVI		
	Absent	0
	Present	38
N-Descriptor		
	N0	0
	N1	25
T-Descriptor		
	T1	0
	T2	17
	T3	58
QuRiS		
	-0.3	0
	-0.2	11
	-0.1	22
	0.0	33
	0.1	44
	0.2	56
	0.3	67
	0.4	78
	0.5	89
	0.6	100

Section 13: Predictive performance of QuRNom.

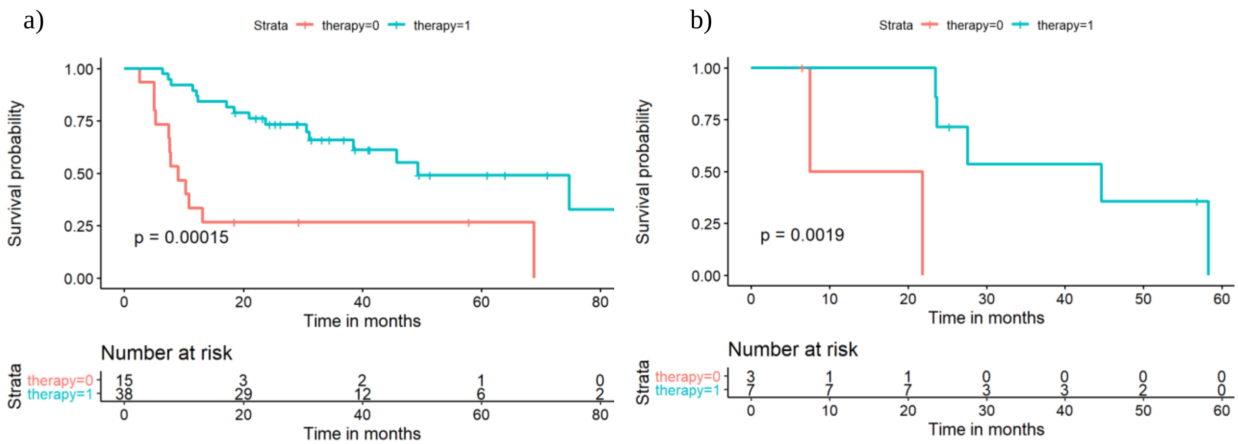


Figure S.8- Subset of patients from developed QuRNom who had predicted three years' disease-free survival less than 20%. a) D_1 and b) D_3 shows that these patients had significantly better survival when received ACT.

Section 14: Radio-Pathomic Analysis: SpaTIL Feature Extraction.

The whole-slide tissue scans for corresponding patients from D1 (N=70) were obtained for extracting spatial tissue infiltrating lymphocytes features (SpaTIL) relating to (i) the spatial architecture of tissue infiltrating lymphocytes (TIL) clusters, (ii) co-localization of clusters of both TILs and cancer nuclei, and (iii) variation in density of TIL clusters across the tissue slide image.

Identification of lymphocytes. The first step in the process was to identify the spatial location of the TILs on digitized H&E stained images. The watershed algorithm was used to automatically detect nuclei. Once the nuclei were detected, a support vector machine using 7 image-derived features related to texture, shape, and color attributes of the segmented nuclei was used to classify the individual nuclei as corresponding either to TILs or to non-TILs.

Quantitative evaluation of spatial arrangement of TILs.

Spatial TIL graph construction. For extraction of SpaTIL features, sets of clusters of proximal TILs and non-TILs were identified. For this purpose, we represented the centroids of each of the individual TILs and non-TILs as nodes of a graph. Each node was connected to others based on a weighted Euclidean norm in which a weighting function favors the connectivity between proximal nodes. Using this process, we generated multiple subgraphs of TILs and non-TILs.

SpaTIL features. The first set of SpaTIL features comprised 20 features related to spatial arrangement of TILs, extracted from the TIL cluster graphs. These features included descriptors such as number of lymphocytes, ratio between area of TIL clusters and area of whole-slide images, and ratio between the number of TILs within the cluster and the cluster area. The second set included the ratio between the non-lymphocyte clusters to the closest lymphocyte clusters, intersection area of lymphocyte and non-lymphocyte, and a value indicating if the nearest cluster neighbor of a specific lymphocyte cluster is either lymphocyte or non-lymphocyte.

These features are listed in Table S4. Each of these features was correlated with developed QuRiS to identify the representation of QuRiS on a whole-slide tissue scans.

Section 15: Radio-Pathomic Analysis: Correlation coefficients between SpaTIL features and QuRiS.

Table S.7 – Pearson Correlation Coefficient

Idx	SpaTIL Features	Correlation Coefficient	Idx	SpaTIL Features	Correlation Coefficient
1	avg of ratio of the nuceli cluster density to its closest lym cluster cluster'	-0.068548217	48	'ratio of number of unique lym nodes to all lym nodes'	-0.116244702

2	'avg of ratio of the nuceli cluster density to its second closest lym cluster density'	-0.150885109	49	'average distance of nuclei to nearest lym'	-0.175162033
3	'avg of ratio of the nuceli cluster density to its third closest lym cluster density'	-0.082589705	50	'lympTotArea'	-0.194999154
4	'avg of ratio of the nuceli cluster density to its first three closest lym cluster mean density'	0.160823937	51	'lymAvgArea'	-0.137579828
5	'std of ratio of the nuceli cluster density to its closest lym cluster cluster'	-0.028989277	52	'lymMedianArea'	-0.008865348
6	'std of ratio of the nuceli cluster density to its second closest lym cluster density'	-0.20681203	53	'lymStdArea'	-0.089769515
7	'std of ratio of the nuceli cluster density to its third closest lym cluster density'	-0.024038741	54	'lymModeArea'	0.058361397
8	'std of ratio of the nuceli cluster density to its first three closest lym cluster mean density'	0.135962339	55	'nonLympTotArea'	0.079192836
9	'var of ratio of the nuceli cluster density to its closest lym cluster cluster'	-0.008784743	56	'nonLymAvgArea'	0.001491959
10	'var of ratio of the nuceli cluster density to its second closest lym cluster density'	-0.197582571	57	'nonLymMedianArea'	0.044094657
11	'var of ratio of the nuceli cluster density to its third closest lym cluster density'	-0.024123586	58	'nonLymStdArea'	-0.078023118
12	'var of ratio of the nuceli cluster density to its first three closest lym cluster mean density'	0.132752781	59	'nonLymModeArea'	-0.08835839
13	'min of ratio of the nuceli cluster density to its closest lym cluster cluster'	-0.04774157	60	'lympTotDensity'	-0.067530353
14	'min of ratio of the nuceli cluster density to its second closest lym cluster density'	-0.194530973	61	'lymAvgDensity'	-0.086611609
15	'min of ratio of the nuceli cluster density to its third closest lym cluster density'	-0.196108574	62	'lymMedianDensity'	-0.064898542
16	'min of ratio of the nuceli cluster density to its first three closest lym cluster mean density'	0.034175957	63	'lymStdDensity'	0.071426038
17	'max of ratio of the nuceli cluster density to its closest lym cluster cluster'	-0.024092555	64	'lymModeDensity'	-0.034241092
18	'max of ratio of the nuceli cluster density to its second closest lym cluster density'	-0.141794394	65	'nonLympTotDensity'	0.107552768
19	'max of ratio of the nuceli cluster density to its third closest lym cluster density'	-0.156591248	66	'nonLymAvgDensity'	-0.007827012
20	'max of ratio of the nuceli cluster density to its first three closest lym cluster mean density'	0.031379749	***67***	***'nonLymMedianDensity'***	***0.220731128***
21	'range of ratio of the nuceli cluster density to its closest lym cluster cluster'	-0.210614368	***68***	***'nonLymStdDensity'***	***0.237694262***
22	'range of ratio of the nuceli cluster density to its second closest lym cluster density'	-0.141450778	69	'nonLymModeDensity'	0.032356646
23	'range of ratio of the nuceli cluster density to its third closest lym cluster density'	0.004059192	70	'intersectedArea'	0.023812444
24	'range of ratio of the nuceli cluster density to its first three closest lym cluster mean density'	-0.014271201	71	'avg1Nearest'	-0.13938205
25	'avg of reciprocal of number of least lym cluster to encompass nuclei cluster'	0.195204313	72	'median1Nearest'	-0.021537739
26	'std of reciprocal of number of least lym cluster to encompass nuclei cluster'	0.223889325	73	'mode1Nearest'	-0.079029197
27	'var of reciprocal of number of least lym cluster to encompass nuclei cluster'	0.16548175	74	'avg2Nearest'	0.108534514
28	'min of reciprocal of number of least lym cluster to encompass nuclei cluster'	-0.212727524	75	'median2Nearest'	0.029155776
29	'max of reciprocal of number of least lym cluster to encompass nuclei cluster'	-0.212670795	76	'mode2Nearest'	0.077753569
30	'range of reciprocal of number of least lym cluster to encompass nuclei cluster'	-0.209706821	77	'avg3Nearest'	0.19617502
31	'avg of area ratio of nuclei cluster to encompassed lym clusters'	-0.115741545	78	'median3Nearest'	0.138335766
32	'std of area ratio of nuclei cluster to encompassed lym clusters'	-0.097028858	79	'mode3Nearest'	0.164061198
33	'var of area ratio of nuclei cluster to encompassed lym clusters'	-0.083277923	80	'avg4Nearest'	0.144128255
34	'min of area ratio of nuclei cluster to encompassed lym clusters'	0.028592185	81	'median4Nearest'	-0.024208296
35	'max of area ratio of nuclei cluster to encompassed lym clusters'	0.055189072	82	'mode4Nearest'	0.153365785

36	'range of area ratio of nuclei cluster to encompassed lym clusters'	0.070868536	83	'avg5Nearest'	-0.061147535
37	'avg of density ratio of nuclei cluster to encompassed lym clusters'	0.05082384	84	'median5Nearest'	-0.082526071
38	'std of density ratio of nuclei cluster to encompassed lym clusters'	0.14941434	85	'mode5Nearest'	-0.06654644
39	'var of density ratio of nuclei cluster to encompassed lym clusters'	0.21056833			
<u>40</u>	***<u>'min of density ratio of nuclei cluster to encompassed lym clusters'</u>***	***<u>0.239931268</u>***			
<u>41</u>	***<u>'max of density ratio of nuclei cluster to encompassed lym clusters'</u>***	***<u>0.247720455</u>***			
42	'range of density ratio of nuclei cluster to encompassed lym clusters'	0.0896348			
43	'1-norm of the vector by ratio of nucleiDegree to LymDegree'	0.141759087			
44	'1-norm of the vector by ratio of nucleiCloseness to LymCloseness'	0.220381499			
45	'1-norm of the vector by nucleiBetweenness minus LymBetweenness'	0.237424695			
46	'1-norm of the vector by ratio of nucleiEigenvector to LymEigenvector'	0.235044918			
47	'1-norm of the vector by ratio of nucleiSumIndex to LymSumIndex'	0.10614462			

Section 16: Radio-Pathomic Analysis: Heatmap for individual prognostic Radiomic Features from QuRiS and SpaTIL.

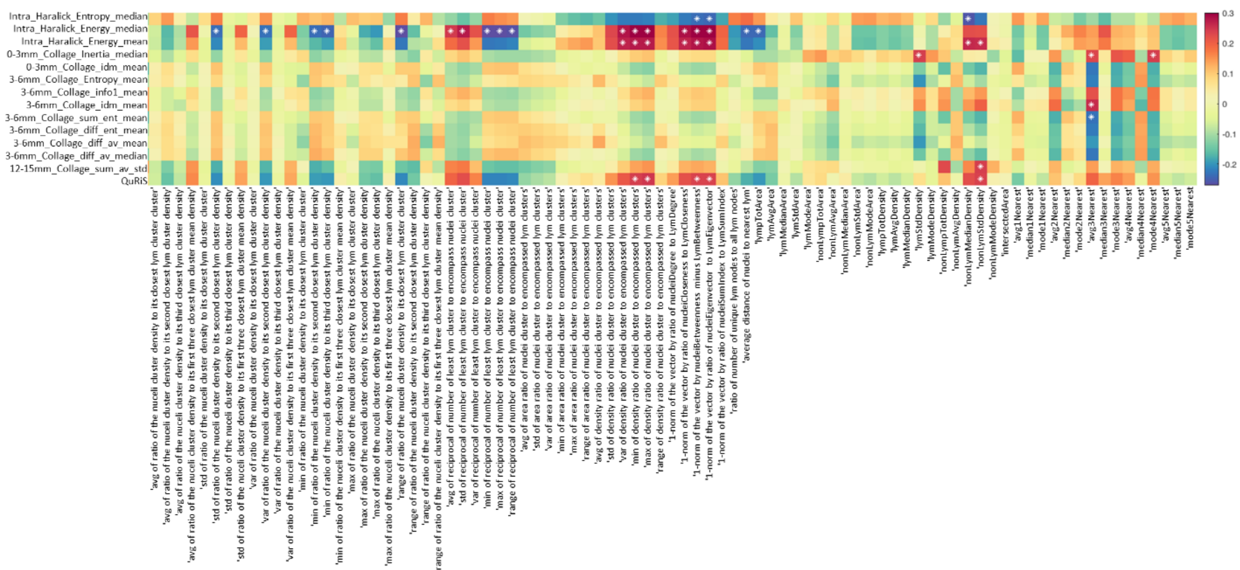


Figure S.9- Pearson correlation coefficients for individual prognostic radiomic features from QuRiS against SpaTIL features. The asterisks represent statistically significant correlation between the features. QuRiS and specifically, intratumoral Haralick features were observed to be correlated with features explaining ratio between cancerous clusters to closest lymphocyte clusters. They were also observed to be correlated with the density of cancerous clusters.

Section 17: Radio-Genomic Analysis: Differentially expressing genes

A total of 22125 genes were reduced to 322 genes using Pearson correlation coefficient between individual genes and developed QuRiS using N=82 samples from D3. The absolute value of rho was restricted to ≥ 0.283 along with $p<0.01$. The top selected genes are listed in appendix-1.

Section 18: Radio-Genomic Analysis: Gene Ontology (GO) analysis

Web-based application, GOView (http://www.webgestalt.org/GOView/) was used to visualize and GO biological processes. GOView enabled comparison of multiple GO processes to identify the common and specific biological themes, by exploring the semantic relationships between the 'parent' and 'child' biological processes. All the biological pathways emerged are shown in sheet-2. After selecting specific biological pathway, gene-set enrichment scores were calculated for each patient using R software version 3.5.1.

Section 19: Radio-Genomic Analysis: Heatmap for individual prognostic Radiomic and biological processes

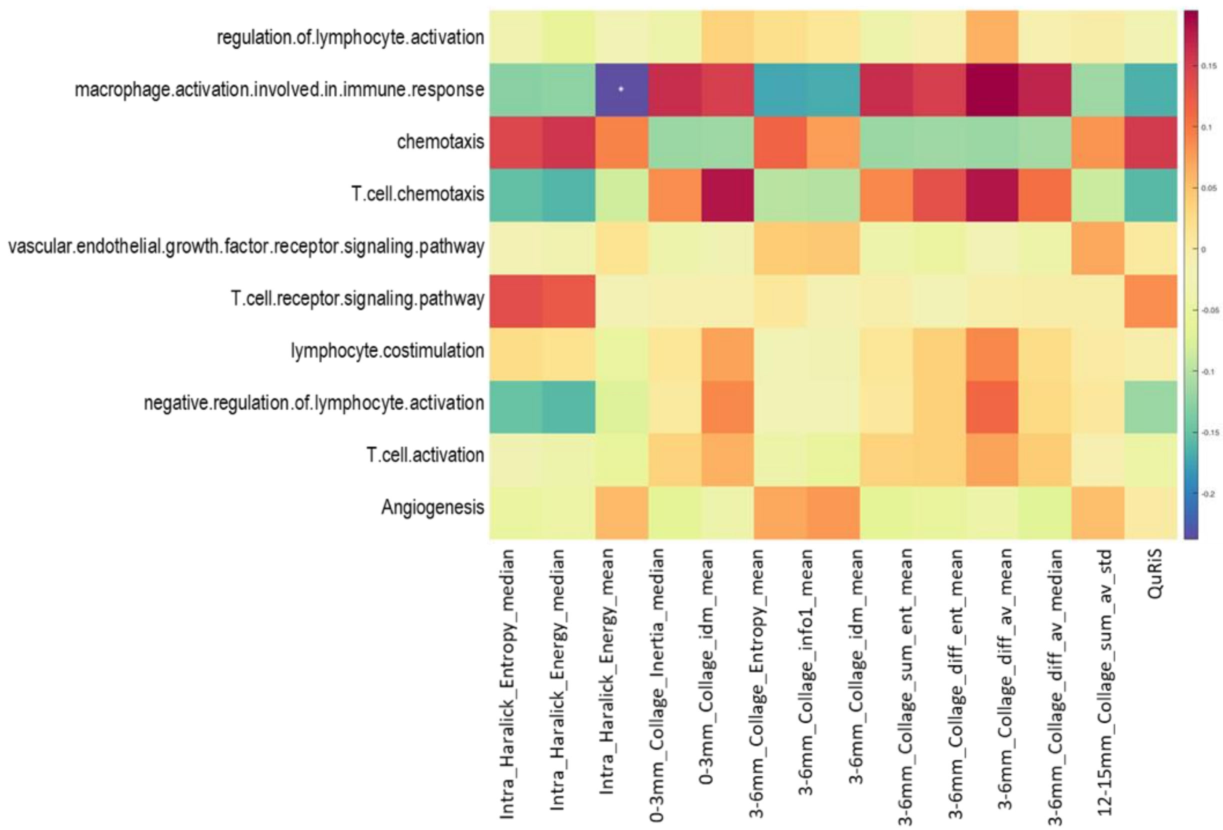


Figure S.10- Pearson's correlation coefficients between gene set enrichment scores of immune related biological processes and top QuRiS Radiomic features. Intratumoral Haralick feature is observed to be inversely correlated with macrophage activation involved in immune response. The developed QuRiS was observed to be weakly correlated with cell chemotaxis. The asterisks represent statistically significant correlation between the features.